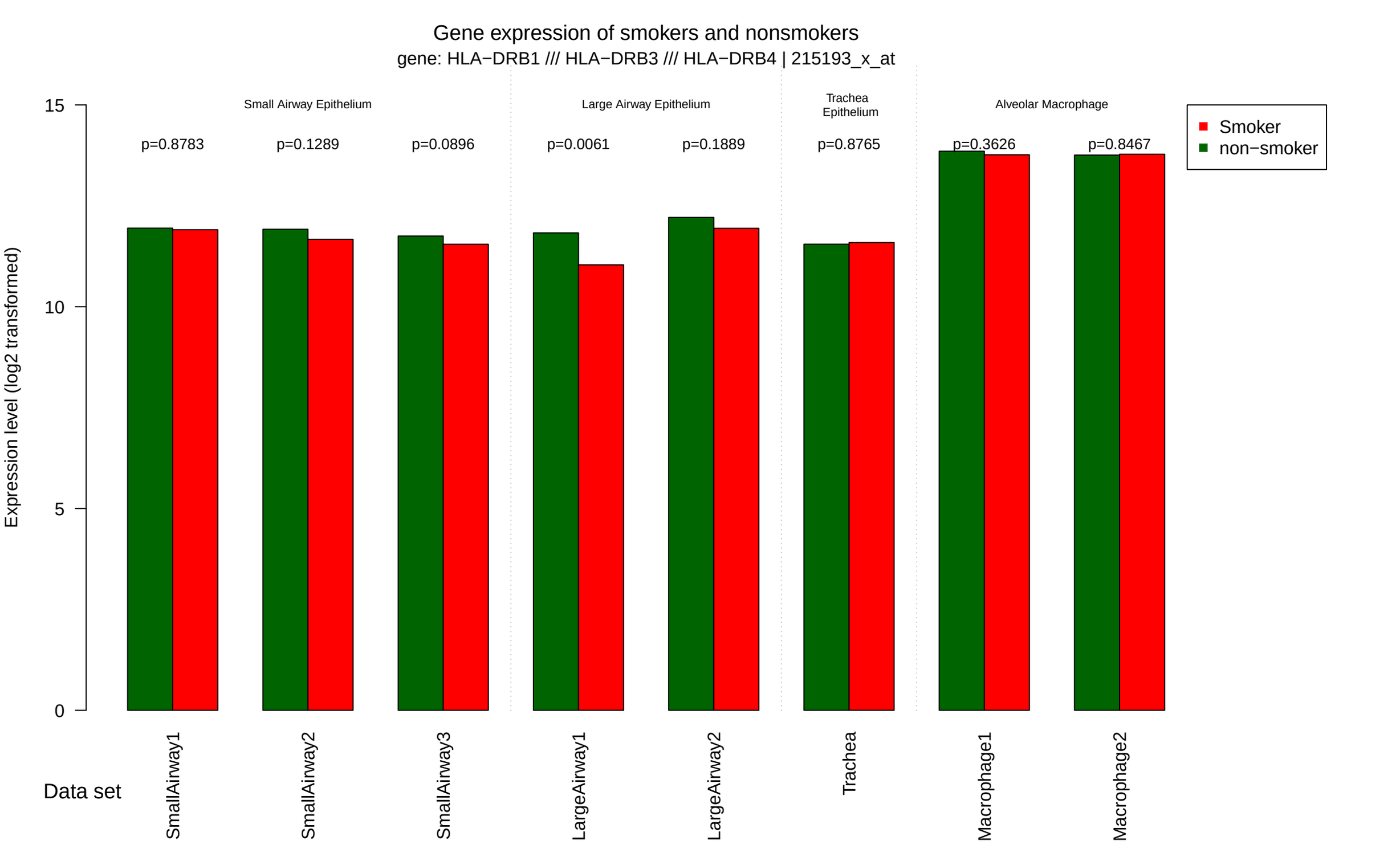
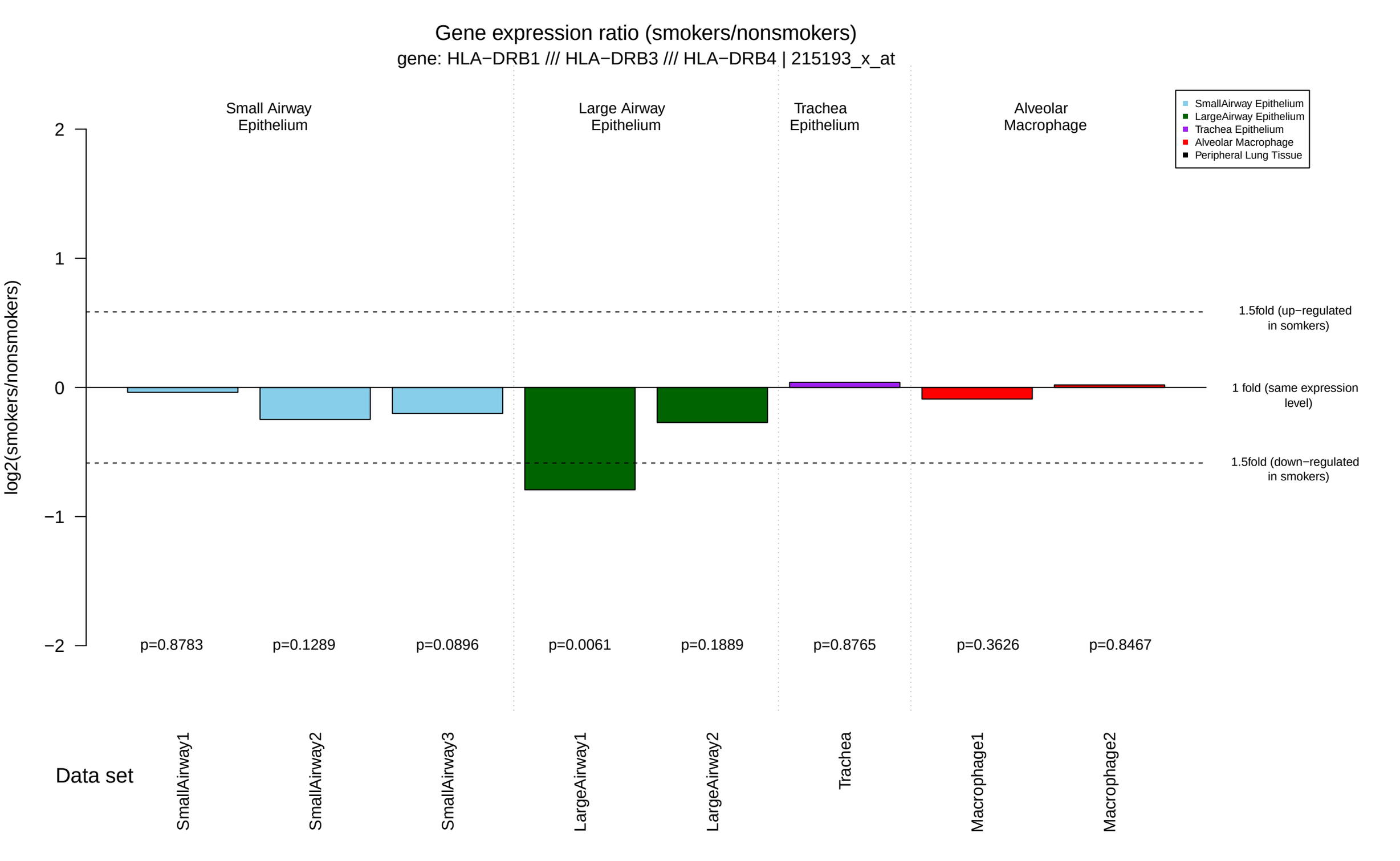
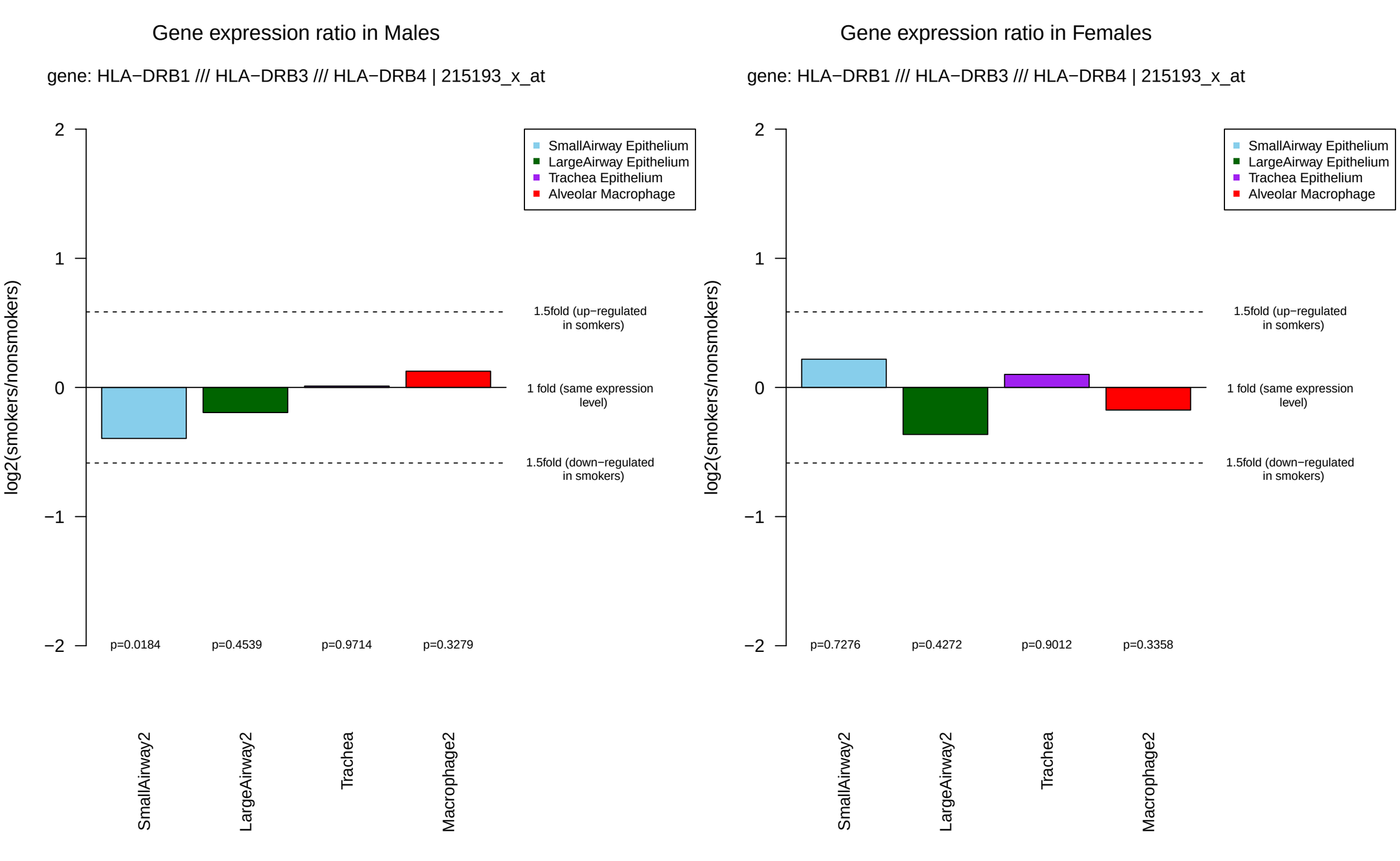
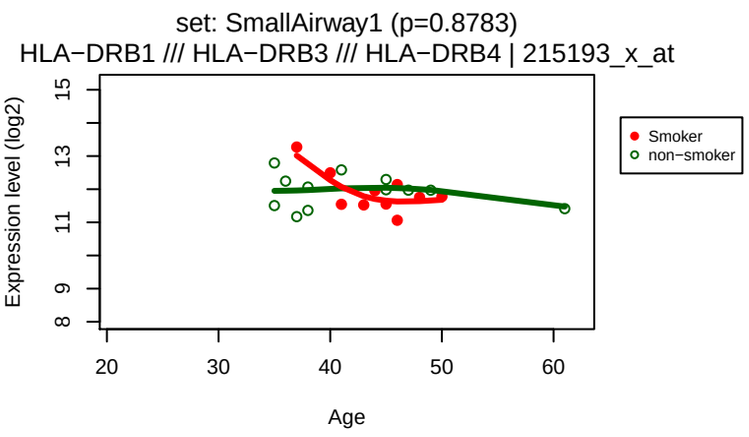
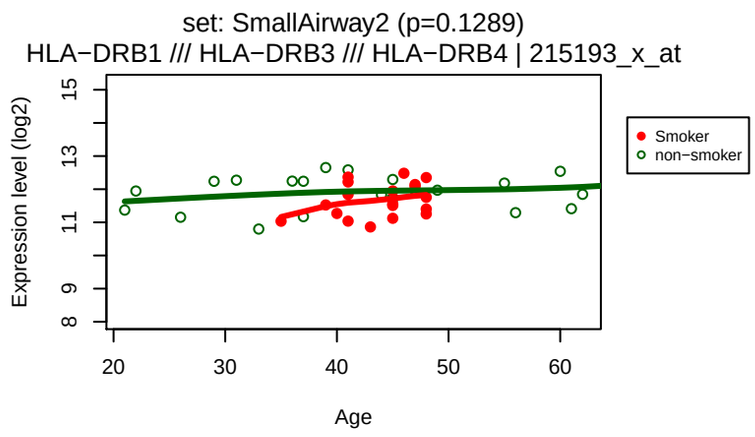
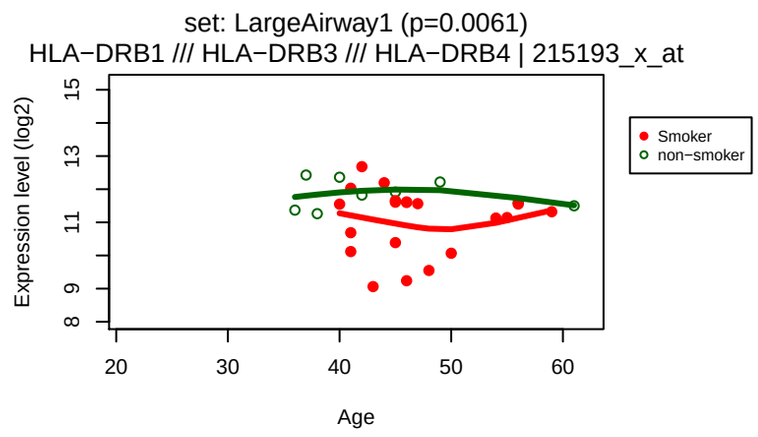

Gene expression of smokers and nonsmokers
gene: HLA−DRB1 /// HLA−DRB3 /// HLA−DRB4 | 215193_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8783
p=0.1289
p=0.0896
p=0.0061
p=0.1889
p=0.8765
p=0.3626
p=0.8467
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

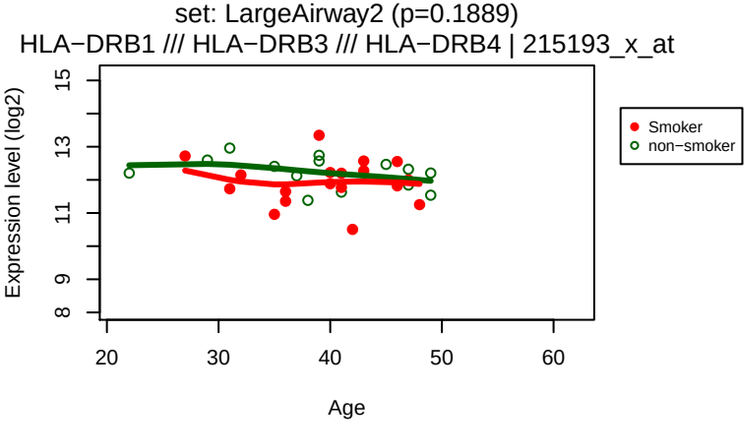

Gene expression ratio (smokers/nonsmokers)
gene: HLA−DRB1 /// HLA−DRB3 /// HLA−DRB4 | 215193_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8783
p=0.1289
p=0.0896
p=0.0061
p=0.1889
p=0.8765
p=0.3626
p=0.8467
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

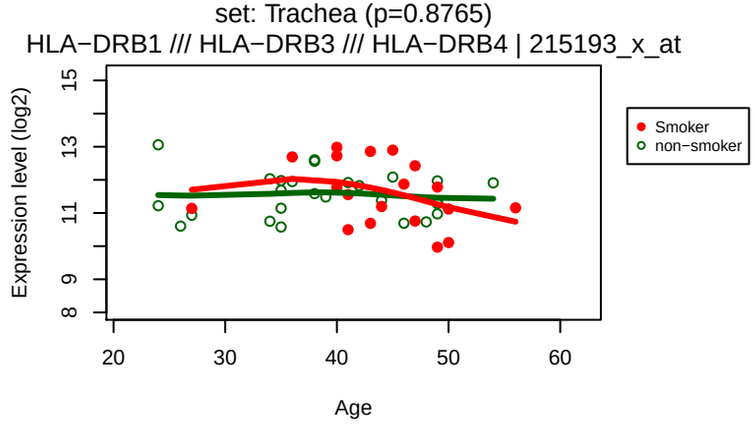

Gene expression ratio in Males
gene: HLA−DRB1 /// HLA−DRB3 /// HLA−DRB4 | 215193_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0184
p=0.4539
p=0.9714
p=0.3279
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−DRB1 /// HLA−DRB3 /// HLA−DRB4 | 215193_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7276
p=0.4272
p=0.9012
p=0.3358
SmallAirway2
LargeAirway2
Trachea
Macrophage2

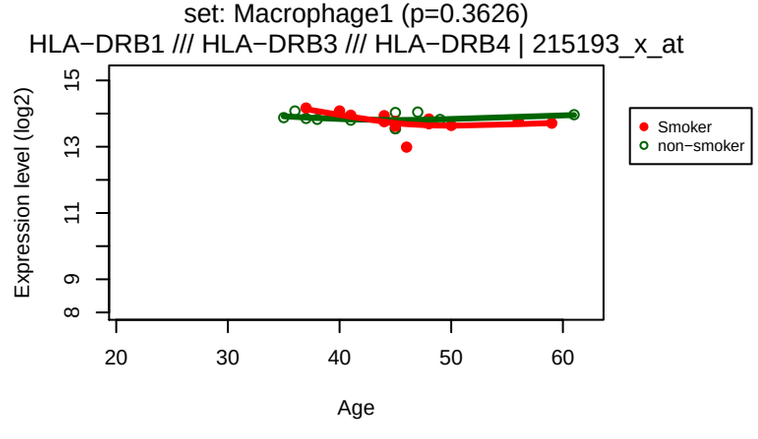

set: SmallAirway1 (p=0.8783)
HLA−DRB1 /// HLA−DRB3 /// HLA−DRB4 | 215193_x_at
Expression level (log2)
Age
Smoker
non−smoker

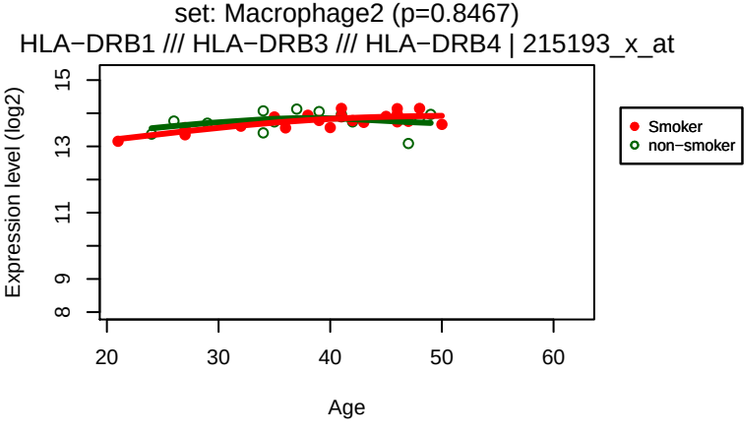

set: SmallAirway2 (p=0.1289)
HLA−DRB1 /// HLA−DRB3 /// HLA−DRB4 | 215193_x_at
Expression level (log2)
Age
Smoker
non−smoker

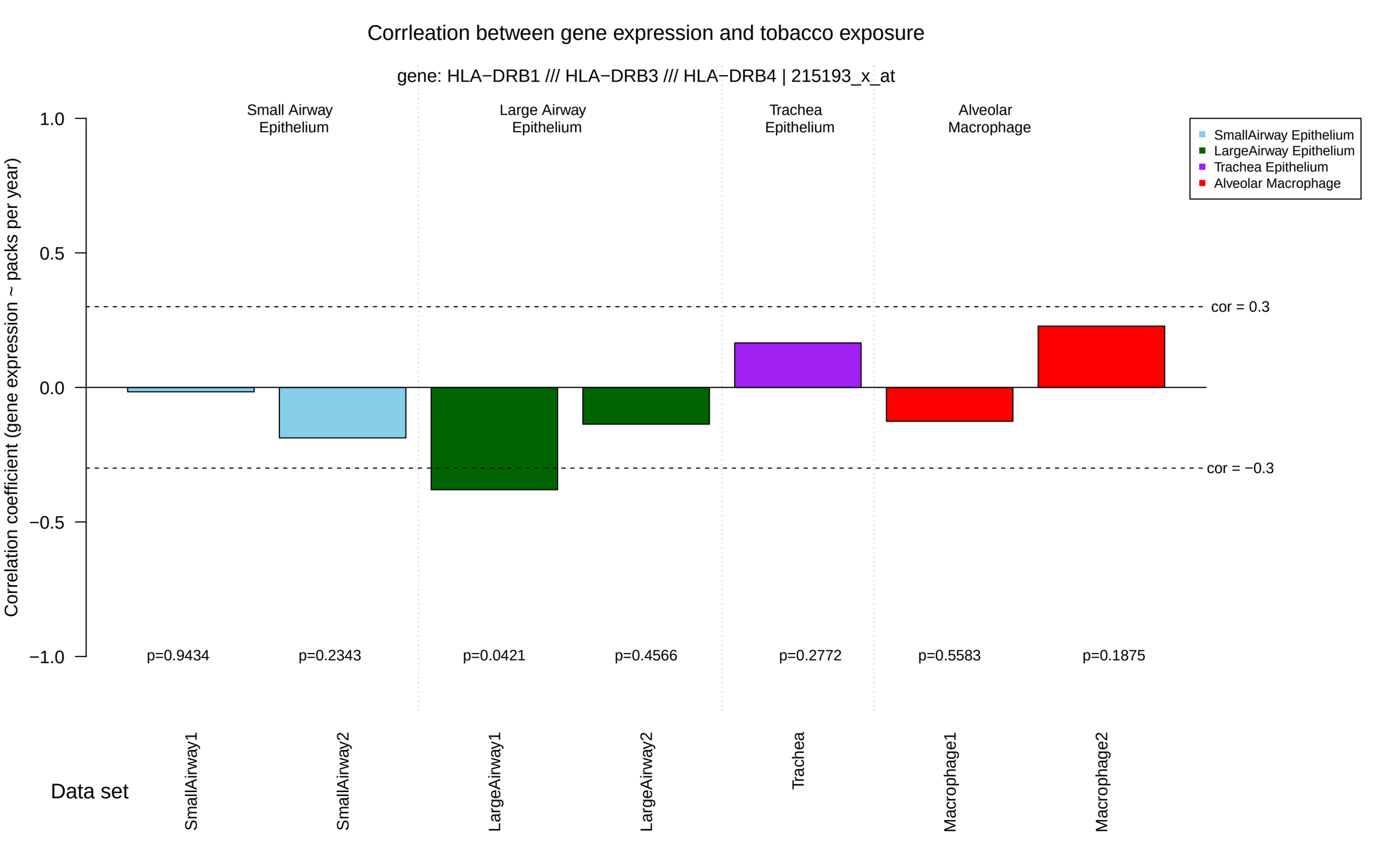

set: LargeAirway1 (p=0.0061)
HLA−DRB1 /// HLA−DRB3 /// HLA−DRB4 | 215193_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1889)
HLA−DRB1 /// HLA−DRB3 /// HLA−DRB4 | 215193_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8765)
HLA−DRB1 /// HLA−DRB3 /// HLA−DRB4 | 215193_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3626)
HLA−DRB1 /// HLA−DRB3 /// HLA−DRB4 | 215193_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8467)
HLA−DRB1 /// HLA−DRB3 /// HLA−DRB4 | 215193_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−DRB1 /// HLA−DRB3 /// HLA−DRB4 | 215193_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9434
p=0.2343
p=0.0421
p=0.4566
p=0.2772
p=0.5583
p=0.1875
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2